

### Objective:

Create a simple AI-based image processing program that can classify images into basic categories. This assignment is meant to test your understanding of image preprocessing and basic machine learning concepts.

### Task Description:

Build a small image classification application using Python. You may use an easy and well-known dataset such as MNIST (handwritten digits) or a small image folder with two classes. A simple pre-trained model or a basic CNN is sufficient.

### Tools & Technologies:

- Python
- NumPy
- OpenCV (optional)
- TensorFlow or PyTorch

### Key Requirements:

- Load images from a dataset
- Resize and normalize images
- Train or use a simple image classification model
- Predict the class of a sample image
- Display the prediction result

### Deliverables:

- Source code (ZIP file or GitHub link)
- README file with steps to run the program
- Short explanation (half to one page) of your approach

### Evaluation Criteria:

- Correct working solution
- Clear and simple code
- Basic understanding of image processing and AI concepts